MILITARY SPECIFICATION

LIGHT, SURGICAL, BRACKET, PORTABLE, BATTERY OPERATED

1. SCOPE

1.1 This specification covers one type of lightweight, battery-operated, portable, field, operating light. Light shall be furnished without batteries or lamps.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-A-200/3	Aluminum Alloy Bar, Rod, Shapes, and Tube Extruded, 2024.
QQ-A-225/6	Aluminum Alloy Bar, Rod, and Wire, Rolled, Drawn, or Cold Finished, 2024.
QQ-A-250/1	Aluminum Alloy, 1100, Plate and Sheet.
QQ-A-250/4	Aluminum Alloy, 2024, Plate and Sheet.
QQ-A-250/8	Aluminum Alloy, 5052, Plate and Sheet.
QQ-B-626	Brass, Leaded and Non-Leaded; Rod, Shapes, Forgings, and Flat Products with Finished Edges (Bar, Flat Wire, and Strips).
QQ-S-763	Steel Bars, Shapes, and Forgings-Corrosion-Resisting.
QQ-W-412	Wire, Steel, Alloy (General Purpose Quality) for Mechanical Springs.
TT-E-529	Enamel, Alkyd, Semi-Gloss.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water-Resistant (for Carton Sealing).

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Military

MIL-C-5541 Chemical Films for Aluminum and Aluminum Alloys.
MIL-L-10547 Liners, Case and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.
MIL-M-14 Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-P-8585 Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity.

STANDARDS

Federal

Fed. Test Method Std. No. 151 Metals; Test Methods.

Fed. Std. No. 595 Colors.

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

DRAWING

Military Medical Supply Agency

2009 Light, Surgical, Bracket, Portable, Battery Operated (revised 22 November 1961).

(Copies of specifications, standards and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BOARD OF FIRE UNDERWRITERS

NBFU No. 56 Standard of the National Board of Fire Underwriters for the Use of Flammable Anesthetics as Recommended by the National Fire Protection Association.

(Application for copies should be addressed to the National Board of Fire Underwriters, 85 John St., N.Y. 38, N.Y., 222 West Adams St., Chicago 6, Ill., and 465 California St., San Francisco 4, Calif.).

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3. REQUIREMENTS

3.1 Material. Material shall be as specified herein. Commercial tolerances shall apply on all structural shapes, tubing, sheets, etc., entering into the construction of the light. All dimensions referring to thickness are nominal gage sizes. Metals shall meet the requirements of their respective specifications when tested as specified in 4.4.1.

3.1.1 Light. The surgical light bracket shall be constructed of the following component parts in the material and quantity specified in the following tabulation:

DRAWING NO. 2009					
Part No.	Sheet No.	Part	Material	Applicable Specification	Quantity
1	2	Receptacle, Sealed Beam	Aluminum-Alloy Sheet	QQ-A-250/1 (1100)	1
2	2	Receptacle Holder	Aluminum-Alloy Bar	QQ-A-200/3 or QQ-A-225/6 (2024 T4)	1
3	2	Ring, Lamp	Aluminum-Alloy Sheet	QQ-A-250/4 (2024 T3)	1
4	2	Lug, Eyebolt	Aluminum-Alloy Sheet	QQ-A-250/8, Condition (5052-H32)	2
5	3	Clamp	Aluminum-Alloy Bar	QQ-A-200/3 or QQ-A-225/6 (2024 T4)	1
6	3	Swivel, Body	Aluminum-Alloy Bar	QQ-A-200/3 or QQ-A-225/6 (2024 T4)	1
7	3	Swivel, End	Aluminum-Alloy Bar	QQ-A-200/3 or QQ-A-225/6 (2024 T4)	1
8	3	Bushing, Locking	Brass, Cadmium Plated	QQ-B-626 Composition 22	1
9	3	Stud, Bearing	Corrosion-Resisting Steel	QQ-S-763 Class 302. condition B or Class 303	1
10	3	Spring	Spring Steel, Oil-Tempered	QQ-W-412 Type I	1
11	3	Handle, Locking	Steel, Cadmium Plated Stud	Commercial	1
			Phenolic Plastic Handle,	MIL-M-14, Type CF1-5	

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as one complete set. Packaged as specified herein shall be shipped

Drawing No. 2009					
Part No.	Sheet No.	Part	Material	Applicable Specification	Quantity
12	3	Washer, Tension	Steel, Cadmium-Plated	Commercial	2
13	3	Washer	Steel, Cadmium-Plated	Commercial	2
14	3	Washer	Steel, Cadmium-Plated	Commercial	1
15	3	Cord, Plug	Electric Cord	Two-Conductor, No. 14, Type SJ or SJO, Black-Rubber-Covered One-Conductor, No. 14, Type SJ or SJO, Green-Rubber-Covered	1
16	3	Cord, Line	Electric Cord	Two-Conductor, No. 14, Type SJ or SJO, Black-Rubber-Covered	1
		Terminal		No. 6-8 Eyelet	2

All parts not given a part number shall be constructed of material and in the quantity shown on the drawing.

3.2 Design and construction. The surgical field light shall consist essentially of a sealed beam receptacle, a swivel bracket, and battery cable. Design and construction shall be as specified herein and as shown on Military Medical Supply Agency Drawing 2009.

3.2.1 Sealed beam receptacle. Sealed beam receptacle shall suitably accommodate a standard 6-volt and 24-volt sealed beam headlight, as shown on the drawing. Finished thickness of receptacle shall be not less than 0.020 inch. The 6-volt lamp socket shall be capable of being pushed aside when the 24-volt sealed beam unit is used. Sealed beam lamps shall not be furnished by the contractor.

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3.2.2 Swivel bracket. Swivel bracket shall permit universal positioning of the housing, and shall have an integral C-clamp suitable for attaching to tubing, up to and including 3/4 inch in diameter.

3.2.3 Line cord. Line cord shall be rubber covered, and shall be fitted with plug-in connectors for the sealed beam headlight at one end, and shall fit into the switch at the other end.

3.2.4 Plug cord. Plug cord shall be rubber covered, and shall fit into the switch at one end and battery clamps at the other end.

3.3 Finish.

3.3.1 Degreasing. All metal surfaces shall be thoroughly cleaned and chemically degreased prior to chemical treatment or plating.

3.3.2 Aluminum parts. All aluminum parts shall be finished with a chemical protective film in accordance with Military Specification MIL-C-5541.

3.3.3 Steel parts. All steel parts, other than corrosion-resisting steel parts, shall be cadmium-plated.

3.3.4 Paint finish. After degreasing, chemical treatment, or plating, a primer coat shall be applied to the exterior surfaces of all metal parts. All cadmium-plated parts to be painted shall be given an acid dip prior to painting. The primer shall be zinc-chromate conforming to Military Specification MIL-P-8585. The primer coat shall form a base for the finishing enamel. The primer coat shall be followed by one coat of baked-on semigloss enamel conforming to Federal Specification TT-E-529 for class B. The color shall be olive drab, No. X24087 (see 6.1) of Federal Standard 595. The resultant finish shall be hard and smooth.

3.4 Identification marking. Each light shall be permanently and legibly marked with the name or registered trademark of the manufacturer. Marking secured by adhesives shall not be employed.

3.5 Service data. The contractor shall enclose, with each unit, two copies each of one or more booklets which contain complete instructions for installation, operation, maintenance, and a list of component parts. Wiring diagrams shall be included for electrical items or components. Each part of the unit other than common fasteners, etc., shall be identified in the service data and parts list by the manufacturer's part number. In addition, the contractor shall mail 12 copies of the service data to the contracting officer for distribution.

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3.6 Electrical parts and wiring methods. All electrical parts and wiring methods shall conform to the requirements of the Underwriters' Laboratories, Inc. The label or listing of the Underwriters' Laboratories, Inc., will be accepted as conforming to this requirement.

3.6.1 In lieu of the label or listing, the contractor may submit prior to delivery independent proof satisfactory to the contracting agency that his equipment conforms to the published standards, including methods of tests, of the Underwriters' Laboratories, Inc.

3.6.1.1 Compliance with Underwriters' Laboratories, Inc., requirements regarding fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his commodity.

3.7 Workmanship. Workmanship shall be first class throughout. All adjustments shall operate easily and smoothly. Light brackets shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with Military Standard MIL-STD-105.

4.2.2 Raw materials. Four ounces of each lot of each metal used in the construction of the light shall be selected for chemical analysis.

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4.3 Examination.. The lights shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests. Tests shall be conducted, to determine compliance with specification requirements.

4.4.1 Chemical composition. Tests for chemical analysis shall be conducted in accordance with method 111.1 or 112.1 of Federal Test Method Standard 151.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One (1) light bracket, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each unit and exterior container, as applicable, for the required level of protection specified in procurement document.

Table I - Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	Not required	6 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP-B-636, type I, class 1. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents. Carton shall be secured.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with Federal Specification PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type I, class 2.

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5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to Military Specification MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix of Federal Specification PPP-B-636. In addition, all seams, including manufacturer's joint, shall be closed by tape conforming to the requirements of Federal Specification PPP-T-76. The tape shall be not less than 2 inches wide. Three strips shall be applied to both top and bottom so that all seams and joints shall be covered their full length. Tape applied over the lengthwise seams shall extend a minimum of 2 inches onto each of the end panels. Tape applied over the end edge seams shall extend a minimum of 2 inches along the side and top, and side and bottom panels.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to Federal Specification PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with Federal Specification PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type I, class 1, fabricated from not less than 275-pound test fiberboard.

5.3 Marking.

5.3.1 Interior packages. Each unit package shall be marked in accordance with Military Standard MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with Military Standard MIL-STD-129. In addition, the letter "L" shall precede the Federal stock number. One space shall be allowed between the letter "L" and the first digit of the stock number. The terms "Stock number" or "FSN" shall not be indicated.

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6 NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.1 Availability of color standards. The color standards of the color referenced in 3.3.4 are available from:

Commanding General
Aberdeen Proving Ground, Maryland
Attention: Coating and Chemical Laboratory

6.1.1 Commercial references. Where components are specified in terms of commercial part numbers, the equal of these parts as produced by other manufacturers may be substituted.

Custodians:
Army. - MD
Navy. - BuMed
Air Force. - AFMSG

Preparing Activity:
Defense Supply Agency. - DM

Review activities:
Army. - MD
Navy. - BuMed
Air Force - AFMSG
Defense Supply Agency. - DM

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET	Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use of the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A
☐ Direct Government Contract ☐ Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD Form 1426

AF Olmsted Sep 63, 150

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DEFENSE SUPPLY AGENCY
DEFENSE MEDICAL SUPPLY CENTER
3RD AVENUE AND 29TH STREET
BROOKLYN, N.Y., 11232

OFFICIAL BUSINESS

POSTAGE FEES PAID
DEFENSE SUPPLY AGENCY

Defense Supply Agency
Defense Medical Supply Center
ATTN: Directorate of Technical Operations (Code T-M2)
3rd Avenue and 29th Street
Brooklyn, N.Y., 11232

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